Appendix: Read Directions — Definition Levers

A *read direction* is an input-side direction of the residual stream that the OV circuits of the selected function-vector heads map onto the task imitation space: the direction the circuit "reads" in order to write the task content it outputs. Given a head h with OV circuit $W_O^h W_V^h \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, its mean task activation $h_A \in \mathbb{R}^{d_{\text{model}}}$, the selected head set H, and the function vector $v_A = \sum_{h \in H} h_A$, a read direction is some vector $r_A \in \mathbb{R}^{d_{\text{model}}}$ whose image under the circuit is aligned with the task content. For GPT-J-6B, $d_{\text{model}} = 4096$ and $d_{\text{head}} = 256$, so each single-head circuit $W_O^h W_V^h$ has rank at most 256 and a kernel of dimension at least 3840.

There is more than one defensible way to make "aligned with the task content" precise, and the choices interact. Rather than committing to a single definition, we state the design space as four levers and sweep over the resulting definitions, selecting among them empirically. Throughout, "target" denotes the vector the read direction should reproduce through the circuit — h_A for a single head, v_A (or its per-prompt analogue v_A^j) for the aggregated constructions. The choice of task-level versus per-prompt target is orthogonal to the levers below and is held fixed within any sweep.

Lever 1 — Optimisation metric: dot product vs. cosine similarity

The first choice is what "the input direction whose image is maximally aligned with the target" means. Let W stand for the relevant circuit (a single head's $W_O^h W_V^h$, or the summed circuit M of Lever 3) and t for the target.

1. **Dot product.** Maximise the inner product of the image with the target,

$$r \;=\; \operatorname*{arg\,max}_{\|z\|=1} \langle Wz,\, t\rangle \quad \propto \quad W^\top t. \tag{1}$$

   The closed form follows from $\langle Wz, t\rangle = \langle z, W^\top t\rangle$. No kernel constraint is needed: $W^\top t$ lies in the row space of W automatically. This objective rewards output *magnitude*: input directions the circuit transmits strongly are favoured even when their image is less purely aligned with t.

2. **Cosine similarity.** Maximise the angle-based alignment of the image with the target,

$$r \;=\; \operatorname*{arg\,max}_{\substack{\|z\|=1 \\ z \,\perp\, \ker(W)}} \cos(Wz,\, t) \quad \propto \quad W^{+} t, \tag{2}$$

   where W^{+} is the Moore–Penrose pseudo-inverse and the kernel constraint makes the maximiser unique (adding any $u \in \ker(W)$ to a maximiser leaves the output unchanged). This objective is indifferent to output magnitude and rewards purity of alignment; the price is that W^{+} scales components by $1/\sigma_i$, so near-zero singular values can dominate the solution. Lever 2 exists to control this.

The two objectives are genuinely different optimisation problems with different solutions: the unit-norm constraint fixes the norm of the *input* z, but the cosine additionally normalises by the norm of the *output* Wz, which still varies with z. The dot product rewards output magnitude along t; the cosine does not.

Lever 2 — Inversion: literal pseudo-inverse vs. τ-truncation (cosine metric only)

This lever applies when Lever 1 selects the cosine metric; the dot-product solution involves no inversion, so the lever collapses there.

1. **Literal pseudo-inverse W^+.** Invert every nonzero singular value. This is the exact maximiser of the cosine objective, but it amplifies whatever the circuit barely transmits: components of t along output directions with small σ_i receive weight $1/\sigma_i$ in the solution, so the read direction can be dominated by directions of near-zero causal throughput.
2. **τ-truncated pseudo-inverse W_τ^+.** Invert only the singular values $\sigma_i \geq \tau$; equivalently, additionally constrain z to be orthogonal to the right-singular subspace with $\sigma_i < \tau$. This trades a small loss in achievable cosine for robustness to directions the circuit does not meaningfully transmit. The threshold τ (equivalently, a rank cut k) is a hyperparameter of the sweep; $\tau \to 0$ recovers the literal pseudo-inverse, and truncation to the top singular direction is the opposite extreme.

(A truncated variant of the dot-product solution — projecting $W^\top t$ onto the top-k right-singular subspace — is definable but is not part of the present crossing.)

Lever 3 — Aggregation across heads: summed circuit vs. per-head-then-sum

The function vector is carried by multiple heads, so a single task-level read direction requires a choice of how the heads combine.

1. **Summed circuit.** Form the summed OV circuit of the selected heads,

$$M \;=\; \sum_{h \in H} W_O^h W_V^h \;\in\; \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}, \tag{3}$$

take the target $t = v_A$ (or v_A^j), and solve the Lever 1 problem once against M. This treats every head as reading the *same* residual-stream vector z, even though the heads sit at different layers — a physical inconsistency accepted for tractability. Note that $\operatorname{rank}(M)$ can reach $d_{\text{head}}\,|H|$, so with $|H| \gtrsim d_{\text{model}}/d_{\text{head}} = 16$ heads M can be full rank; the kernel constraint of eq. (2) is then vacuous and τ-truncation (Lever 2) becomes the only regulariser.

2. **Per-head, then sum and normalise.** For each head $h \in H$, solve the Lever 1 problem for that head's own circuit $W_O^h W_V^h$ against that head's own mean activation h_A, giving a per-head read direction r_A^h; then aggregate:

$$r_A \;=\; \frac{\sum_{h \in H} r_A^h}{\left\|\sum_{h \in H} r_A^h\right\|}. \tag{4}$$

   This respects each head's own input geometry but loses cross-head interactions: no head can compensate for another's blind spot, unlike in the summed circuit.

   Sub-choice (3′). Each r_A^h is unit-norm by construction, so the plain sum in eq. (4) weights heads equally. An alternative weights each head by the magnitude of its unnormalised solution — $\|(W_O^h W_V^h)^{+} h_A\|$ under the cosine metric, or $\|(W_O^h W_V^h)^{\top} h_A\|$ under the dot product — before summing.

Lever 4 — Final normalisation: unit norm or natural magnitude

The last choice is whether the final r_A is unit-normalised.

1. **Unit norm.** r_A is a pure direction; any intervention strength is delegated entirely to the steering-coefficient sweep over α.
2. **Natural magnitude.** Keep the norm the construction produces — $\|W^{+}t\|$ under the cosine metric, $\|W^{\top}t\|$ under the dot product, or the norm of the per-head sum in Lever 3.2. These magnitudes carry meaning (for instance, $\|M^{+}v_A\|$ is the input scale required to produce a v_A-sized output through M), and they differ across tasks and heads, so this option changes the relative dosing across tasks in any comparison at fixed α.

This lever is immaterial for purely correlational uses of read directions (the cosine of r_A against residual activations is scale-invariant); it matters for causal uses — injection, ablation, or projection-removal at a fixed intervention strength.

The resulting crossing

Under the dot-product metric Lever 2 collapses, giving $1 \times 2 \times 2 = 4$ definitions; under the cosine metric the crossing is $\{$literal, τ-truncated over a grid of $\tau\} \times 2 \times 2$. Two points of the crossing correspond to the definitions used in the main text: the single-head read direction r_A^h (cosine, literal pseudo-inverse, per-head, unit norm) and the summed-circuit read direction $r_A \propto M^{+}v_A$ (cosine, literal pseudo-inverse, summed circuit, unit norm). The transpose formula $r \propto W^{\top}t$, sometimes proposed as the solution of the cosine problem, is in fact the solution of the dot-product problem; within this crossing it is a legitimate lever point rather than an approximation.

Finally, the criterion by which the best definition is selected is itself a definitional choice — correlational alignment with residual activations at early tokens, or the causal effect of intervening along r_A, and in either case under which prompt distribution — and must be fixed before the sweep is run.